

## Lecture 5 - Sep 9

### Quick Review

$$\begin{aligned}M &= \{n_1 a_1, \dots, n_k a_k\} \\n &= n_1 + \dots + n_k \\r &\leq \min\{n_1, \dots, n_k\} \\ \# \text{ } r\text{-permutations of } M &= \underbrace{k * k * \dots * k}_{r \text{ times}} = k^r \\ \# \text{ permutations of } M &= \frac{n!}{n_1! \dots n_k!}\end{aligned}$$

### Example

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$$M = \{3 * a, 4 * b, 5 * c\}$$

### Answer

$$\begin{aligned}n &> r = 11 > \min\{3, 4, 5\} \\M_a &= \{2 * a, 4 * b, 5 * c\} \\M_b &= \{3 * a, 3 * b, 5 * c\} \\M_c &= \{3 * a, 4 * b, 4 * c\}\end{aligned}$$

Therefore the answer is

$$\boxed{\frac{11!}{2!4!5!} + \frac{11!}{3!3!5!} + \frac{11!}{3!4!4!}}$$

### Multiset Combinations

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$$\begin{aligned}M &= \{n_1 * a_1, n_2 * a_2, \dots, n_k * a_k\} \\r &\leq \min\{n_1, \dots, n_k\} \\n &= n_1 + \dots + n_k\end{aligned}$$

#  $r$ -combinations of  $M$  = # non--integral solutions to  $x_1 + \dots + x_k = r, x_i \geq 0 =$

$$\# \text{ permutations of } M = \{r * *, (k-1) * 1\}$$

$$= \binom{r+k-1}{r} = \binom{r+k-1}{k-1}$$

### Example

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8 varieties of bagels

Box of dozen bagels (12 bagels)

$$M = \{a * \infty, b * \infty, \dots\}$$

Define  $x_i$  = # bagels of type  $i$

$$x_1 + x_2 + \cdots + x_8 = 12 \rightarrow \binom{12 + 8 - 1}{8}$$

## Example

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What is the # of non-decreasing sequences of length  $r$ , whose terms are taken from  $1, \dots, k$  (multiple allowed, multiset)?

### Answer without the first condition

#  $r$ -permutations of  $M = \{\infty * 1, \dots, \infty * k\}$  is  $k_r$

### Answer

Let  $M = \{\infty * 1, \dots, \infty * k\}$

Then we can construct a non-decreasing sequence as follows:

1. Select  $r$  numbers from  $M$
2. Arrange the  $r$  chosen #'s into a non-decreasing order

By the multiplication principle, # of non-decreasing sequences is

# ways to carry out step (1) \* # ways to carry out step (2)

There is only 1 way to carry out step (2): all distinct elements have one order compared to the others, the order of equal elements don't matter since it's multiset

The # ways to carry out step (1) is the # of  $r$ -combinations of  $M$ , therefore the answer is:

$$\boxed{\binom{r + k - 1}{r} * 1}$$

## Example

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Find the # non-negative integral solution to

$$\begin{cases} x_1 + x_2 + \cdots + x_k = 100 \\ x_1 \leq 5, x_2 \geq 0, \dots, x_{20} \geq 0 \end{cases}$$

### Answer

# non-negative integral solution to

$$x_1 + \cdots + x_{20} = 100$$

$$x_1 \geq 0, x_2 \geq 0, \dots, x_{20} \geq 0$$

minus

# non-negative integral solutions to

$$\begin{aligned}x_1 + \cdots + x_k &= 100 \\ x_1 &\geq 6, x_2 \geq 0, \dots, x_{20} \geq 0\end{aligned}$$

Therefore, the answer is

$$= \binom{100 + 20 - 1}{100} - \binom{(100 - 6) + 20 - 1}{100 - 6}$$

## Chapter 3 - The pidgeonhole Principle

This is an *important but elementary* combinatorial principle that can be used to solve a variety of interesting problem.

### PP (Simple form)

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If  $n + 1$  or more objects are placed into  $n$  boxes, thne same box msut contain at least two of the same object.

#### Proof

Suppose each box contains at most one of the objects. Then we have at most  $n$  objects in total, a contradiction  $\square$ .

*Remark 1:* Neither the statement nor the proof gives any help in finding which box contains two or more of the objects.

#### Example

There are  $n$  couples. How many of the  $2n$  people must be chosen to guarantee that a couple is selected

#### ANSWER

$n + 1$ , by pidgeonhole principle

*Remark 2:* If  $n$  objects are plaecd into  $n$  boxes and no box is empty, then each box contains exactly one object

*Remark 3:* If  $n$  objects are placed into  $n$  boxes adn no box contains two or more of the objects, then each box contains exactly one object.

*Remark 4:*

Let  $X, Y$  be sets and  $f : X \rightarrow Y$

- a. If  $|X| > |Y|$ , then  $f$  is not 1-1
- b. If  $|X| = |Y|$  and  $f$  is onto, then  $f$  must be 1-1 (also meaning  $f$  is a bijection).
- c. If  $|X| = |Y|$  and  $f$  is 1-1, then  $f$  must also be onto (also meaning  $f$  is a bijection).

## Example

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Prove that given any 2026 integers  $a_1, \dots, a_{2026} \in \mathbb{Z}$ .

$$\begin{array}{l} i \neq j, 2025 \nmid a_i - a_j \\ a_i - a_j \text{ is divisible by } 2025 \end{array}$$

To apply *PP* we must know the pigeons and boxes.

**Boxes:** There are at most 2025 different remainders after dividing by 2025:

$$0, 1, \dots, 2024$$

**Pigeons:** We want to place all  $a_i - a_j$  into all the possible remainders (except 0).

$$\begin{array}{l} r_1 = a_1 \mod 2025 \rightarrow \\ a_1 = c_1 2025 + r_1 \\ a_2 = c_2 2025 + r_2 \\ \dots \\ a_{2026} = c_{2026} 2025 + r_{2026} \end{array}$$

$r_i \in \{0, 1, \dots, 2024\}$  For any  $i$ .

$r_1, \dots, r_{2026}$  pigeons

Therefore,  $r_i = r_j, i \neq j$

Then

$$\begin{array}{l} a_i - a_j = c_i * 2025 + r_i - (c_j * 2025 + r_j) \\ = (c_i - c_j) * 2025 + 0 \end{array}$$

so  $2025 \mid a_i - a_j$ , as desired  $\square$ .